

Chapter 3: Waves & Electromagnetism

Wave Properties, Sound, Optics, Electric Fields & Magnetic Induction

3.1 Wave Motion & Sound

A wave transports energy without transporting matter. The fundamental wave equation links wave speed v , frequency f , and wavelength λ : $v = f \lambda$.

Sound waves are longitudinal mechanical waves requiring a medium. The speed of sound in air depends on temperature and density. The Doppler Effect describes the shift in frequency heard by a moving observer relative to a source.

3.2 Geometric & Wave Optics

Optics explores the behavior of light. Reflection follows the Law of Reflection ($\theta_i = \theta_r$), while refraction is governed by Snell's Law:

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2)$$

Lenses and mirrors follow the thin-lens equation: $\frac{1}{f} = \frac{1}{d_o} + \frac{1}{d_i}$.

3.3 Electrostatics & Electromagnetism

- **Coulomb's Law:** $F = k \frac{|q_1 q_2|}{r^2}$, describing electrostatic force between point charges.
- **Ohm's Law:** $V = IR$, relating potential difference V , current I , and resistance R .
- **Magnetic Force & Induction:** Moving charges experience magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$. Faraday's Law of Induction states that a changing magnetic flux induces an electromotive force (EMF).